AI writes code that calls functions that don’t exist.

I built a tool that catches it. **Deterministically.**

FixProve — free, live beta

The code looks right.
It compiles.
It passes a quick review.

Then a method that was
never real gets called —
and it fails at runtime,
or silently does the wrong thing.

This is the everyday
failure mode of
AI-assisted coding,
not an edge case.

The scale

USENIX Security 2025 — 'We Have a Package for You!' (Spracklen et al.)

— 2.23 million code samples, 16 LLMs, Python + JavaScript

— 205,474 unique non-existent package names invented

— Hallucination averaged 21.7% for open-source models, 5.2%+ for commercial ones

This is a measured, published problem — not an opinion.

Rust and Go catch this at build time — their compilers reject calls to things that don't exist before you ship.

Python and JavaScript don't have that safety net.

That's the gap. It's exactly where FixProve works — and exactly why it doesn't bother with Rust or Go.

What FixProve does

Verifies AI-generated code against your project's actually installed dependencies.

Flags calls to functions, methods, classes, and API surfaces that aren't really importable or callable in your environment.

Deterministic static analysis. No LLM in the loop.

You can't reliably check AI code with more AI — if a model can hallucinate the code, it can hallucinate the review of the code too.

FixProve breaks that loop: same input → same output, every run.

A fact check against your real environment, not a second opinion.

How it ships today

- MIT-licensed CLI on PyPI and npm — fixprove, v0.1.10. Install and run it today, free.

- A GitHub App that posts a blocking PR status check in CI — currently internal-only while an independent legal review is in progress.

- Backend on Cloudflare Workers + KV. Fully static site on Cloudflare.

How it’s built

- Every non-trivial change traces to a requirement and a test
- Architectural decisions logged as they’re made
- A deliberate adversarial stage — try to break it before calling it done
- OIDC trusted publishing to PyPI/npm — no long-lived tokens
- Full CI test suites across the Python resolver, the TS/JS resolver, and the Cloudflare Worker

The honest part

I build closely with AI tools myself —
that's exactly why I hold this to a higher bar.

I'm also actively engaging independent Danish law firms for a real compliance review before wider rollout.

That review is in progress, not finished — nothing here is endorsed or certified yet.

FixProve is free, live, and in public beta.
Pre-revenue — no paid tier,
no customer numbers to cite.
Just a tool I think should exist.

Try it: pip install fixprove or npm install fixprove

fixprove.dev

If you ship AI-assisted Python or TypeScript:
how do you catch invented APIs today?
I read every reply.

Yehor Kaliberda — building FixProve solo, Aarhus